



Fig. 3: Formats supported by the camera



Fig. 5: Eyes detection



Fig. 4: Face detection



Fig. 6: Upper body detection

ner and size of a bounding box. Input image *I* must be a gray scale or true colour (RGB) image.

3. The third step is:

```
insertObjectAnnotation(I,'rectangle',
    Position,Label)
```

It inserts rectangles and corresponding labels at the location indicated by the position matrix. The position input must be an M-by-4 matrix, where each row (M) specifies a rectangle as a four-element vector (x, y, width and height). Elements x and y indicate the upper-left corner of the rectangle, and the width and height specify the size.

Testing

To test this program, follow the steps given below:

1. Install MATLAB version R2012a or higher version in your system. Launch it from desktop shortcut. You will see a blank command window.
2. Download the source folder, either from EFY DVD or from website, as given in EFY Note.
3. Check the device ID, as shown in Fig. 2, and write the device ID number in the source code.
4. Run the program. A GUI will appear, as shown in Fig. 1.
5. Click on Start button to initialise camera settings.
6. Next, click on Face button and the camera will detect the face. Your face will be detected and displayed on the right side of the screen (Fig. 4). Similarly, you can also detect your

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

eyes and upper body by clicking on the respective buttons. The output screenshots for the detected eyes and upper body are shown in Fig. 5 and Fig. 6, respectively. Remember to click Stop button first to stop the previous process in order to detect other two features.

7. To stop, click Stop button. ●

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